

CYBERSECURITY IN INDIA

2025–2026 Threat Landscape & Best Practices

DOCUMENT CONTROL & OVERVIEW

This briefing examines the rapid escalation of cyber threats across India's digital ecosystem covering 2025 and 2026. Leveraging telemetry and reports from *MeitY*, *CERT-In*, *CSIRT-Fin*, *SISA*, and *Seqrite Labs*, it details the state of national cyber resilience. It analyzes the emergence of AI-powered attacks, outlines recent high-profile breaches, and provides practical precautions and mandatory regulatory guidelines for enterprise defense.

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1. EXECUTIVE SUMMARY: THE STATE OF CYBERSPACE IN INDIA

India's aggressive digital adoption has unlocked significant economic efficiencies but has simultaneously expanded its systemic cyber vulnerability. In 2025, the Indian Computer Emergency Response Team (CERT-In) reported and tracked a staggering **29,44,248 (29.44 lakh) cybersecurity incidents**. This represents an **85.4% escalation** from the 15,92,917 incidents recorded in 2023, signaling a highly volatile threat environment. Parallel enterprise-level endpoint data from Seqrite Labs documented over **265.52 million malware detections** during this period, averaging a crushing **505 cyber attacks every minute** across the nation.

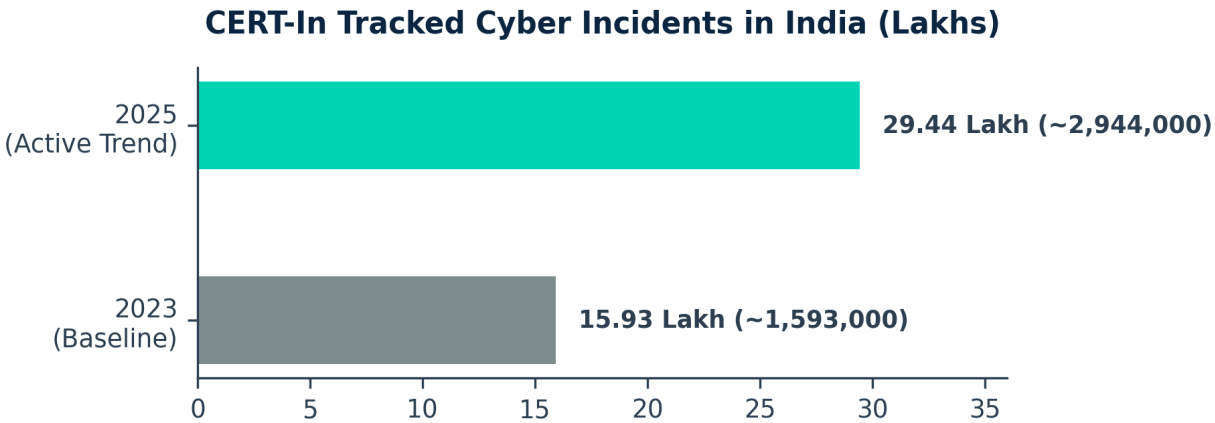


Figure 1: Exponential growth of tracked cybersecurity incidents in India (CERT-In).

The Emergence of AI Asymmetry

A central finding of the joint *Digital Threat Report 2025–26* released by MeitY, CERT-In, CSIRT-Fin, and SISA is the rapid crystallization of **AI Asymmetry**. This refers to the growing technical and operational imbalance where cybercriminals leverage generative AI to develop polymorphic malware, voice-cloned deepfakes, and automated exploit scanners at machine speed, far outpacing the defensive and regulatory capabilities of organizations. The consequence of AI Asymmetry is the complete compression of the interval between threat discovery and operational exploitation—collapsing from years to mere weeks or days. Furthermore, modern attacks no longer resemble typical perimeter penetrations; instead, they hijack, manipulate, and mimic legitimate user sessions, making detection extremely difficult.

2. RECENT CYBERATTACKS IN INDIA (2025–2026)

High-profile, verified security incidents across 2025 and 2026 illustrate that no sector—critical infrastructure, healthcare, government, or finance—is immune. Key case studies highlight dominant vectors:

Operation Sindoor (May 2025): A nationwide coordinated hybrid cyber campaign. Pakistan-aligned hacktivists and state-linked groups targeted critical assets. This included a 19-hour continuous DDoS attack on the President's official website, over 200,000 probing/scanned attempts on the national power grid, and targeted disruptions against NIC data centers.

Delhi Hospitals Server Hack (June 2025): Sant Parmanand Hospital and NKS Super Speciality Hospital reported overnight server hacks. Patient data, administrative files, and financial records were compromised. OPD/IPD digital operations were paralyzed, forcing staff to revert entirely to manual paper workflows.

Nippon India Mutual Fund Portal Outage (April 2025): Nippon Life India Asset Management suffered a severe cyberattack. To isolate and investigate the intrusion, customer-facing systems (including the main website, portal login, and mobile application) were completely shut down for **12 consecutive days**, disrupting digital investor transactions.

Angel One AWS Cloud Compromise (February 2025): One of India's largest stockbroking platforms detected unauthorized access to AWS-hosted cloud environments. To contain the breach, the firm instantly rotated cloud and app credentials and isolated assets. Telemetry from DSCI shows that **62% of all detections** now occur in cloud assets, driven by misconfigured buckets and overly permissive IAM.

Tata Technologies Ransomware (January 2025): The company isolated some internal IT systems due to a ransomware attack. In March 2025, the Hunters International ransomware syndicate claimed responsibility, advertising the alleged theft of **1.4 TB of corporate data** on dark web leak markets.

Operational Sector Vulnerability Framework

Sector	Primary Attack Types	CIA Risk	Key Operational Impacts
BFSI	Cloud compromise, vendor portal malware, DDoS floods	C + A	Nippon Mut. Fund (12-day portal outage), ICICI Bank third-party vendor compromise.
Healthcare	Ransomware extortion, system hacks, data database leaks	C + A	Delhi Hospitals (Sant Parmanand & NKS Super Speciality) OPD/IPD digital workflows shut down.
Govt & Infra	Coordinated DDoS campaigns, web defacements, IT/OT probing	A + I	Operation Sindoor: 19-hr DDoS on President's site, 200k power grid probing attempts, BSNL DDoS.
Defence	APT campaigns, sideloaded DLLs, open-source RATs	C	Ongoing campaigns by state-linked threat actors targeting tactical communication networks.

Table 1: Operational impact and risk classification of cyber attacks across key Indian sectors.

3. EMERGING THREAT VECTORS IN 2026

According to 2026 threat telemetry, attack patterns have pivoted away from raw brute-forcing toward three main domains:

AI-Generated Phishing & Deepfake BEC: Traditional email security training is falling short against voice-cloning and video deepfakes. Attackers mimic CEOs and CFOs over voice calls or virtual meetings to authorize rapid financial transfers or harvest high-privileged administrative credentials.

Supply Chain Multiplying Attacks: Highly secure enterprise perimeters are bypassed by attacking smaller, trusted third-party vendor portals. This is illustrated by the Bashe ransomware group's campaign targeting ICICI Bank's vendor access point, and the Sinobi ransomware group targeting IT Managed Service Providers (MSPs) to access customer virtual machines.

Unified Payments Interface (UPI) Fraud: With UPI processing over 15 billion transactions monthly, financial fraud has scaled massively. MHA data showed 9.42 lakh SIM cards blocked for fraud. Total losses from investment scams, fake stock platforms, and digital arrests are projected to reach ■1.2 lakh crore.

4. ESSENTIAL PRECAUTIONS & COMPLIANCE DIRECTIVES

To mitigate these sophisticated, machine-speed threats, Indian enterprises and individuals must enforce a rigorous defense framework and align with strict cyber hygiene standards:

Multi-Factor Authentication (MFA): Enforce hardware-based or application MFA across all user logins, cloud consoles, and virtual private networks. Never rely on passwords alone.

Zero Trust Network Architecture: Micro-segment corporate networks. Ensure that if a VPN or virtual machine (like Hyper-V) is compromised, the attacker cannot pivot laterally across the rest of the infrastructure.

Immutable Backups: Maintain air-gapped, encrypted, and immutable backups. Test restoration processes quarterly to ensure recovery from destructive ransomware without funding attackers.

Mobile Device Cyber Hygiene: Keep operating systems and applications updated on auto. Disable Bluetooth, Wi-Fi, and Near-Field Communication (NFC) when not in use. Avoid connecting to public, unsecured Wi-Fi without a VPN.

MANDATORY REGULATORY DIRECTIVES IN INDIA

- **Mandatory 6-Hour Reporting Window:** Under CERT-In directions, all organizations operating in India must report cyber security incidents to the national agency within **6 hours** of detection.
- **DPDP Act 2023 Compliance:** Personal data breaches require immediate notifications to the Data Protection Board of India and affected data principals. Heavy financial penalties apply for poor data governance.
- **National Cybercrime Portal & Helpline:** Victims of active financial fraud must report immediately to the **National Helpline 1930** or submit details to *cybercrime.gov.in* to freeze stolen funds in transit.